

Second-Order Linear Homogeneous Equations



The auxiliary equation

- 1 Write the auxiliary (characteristic) equation for $y'' - 5y' + 6y = 0$.
- 2 Write and simplify the auxiliary equation for $2y'' + 4y' - 6y = 0$.
- 3 Explain why substituting $y = e^{rx}$ into $ay'' + by' + cy = 0$ produces the auxiliary equation $ar^2 + br + c = 0$.

Distinct real roots

- 4 Solve $y'' - 5y' + 6y = 0$.
- 5 Solve $y'' + 2y' - 3y = 0$, $y(0) = 1$, $y'(0) = 1$.
- 6 Solve $y'' - y' - 6y = 0$.
- 7 Solve $y'' - 9y = 0$ (note this factors as a difference of squares in r).

Repeated roots

- 8 Solve $y'' - 6y' + 9y = 0$.
- 9 Solve $y'' + 4y' + 4y = 0$, $y(0) = 2$, $y'(0) = -1$.
- 10 Explain why, for a repeated root r , $y_2 = xe^{rx}$ (not just a second copy of e^{rx}) is needed to build the general solution.

Complex roots

- 11 Solve $y'' + 4y = 0$.
- 12 Solve $y'' - 2y' + 5y = 0$.
- 13 Solve $y'' + 9y = 0$, $y(0) = 2$, $y'(0) = 3$.
- 14 Solve $y'' + 2y' + 2y = 0$.

Classifying without fully solving

- 15 Without solving, state whether $y'' + 6y' + 5y = 0$ has real, repeated, or complex roots, using the discriminant $b^2 - 4ac$ of the auxiliary equation.
- 16 Without solving, classify the roots of $y'' + 10y' + 25y = 0$.
- 17 Without solving, classify the roots of $y'' + 2y' + 10y = 0$.

Application: a damped spring

- 18 A mass–spring–damper system satisfies $y'' + 4y' + 5y = 0$. Find the general solution and classify the damping (over/critically/under-damped).
- 19 Solve $y'' + 6y' + 9y = 0$, $y(0) = 1$, $y'(0) = 0$, and classify the damping.
- 20 Solve $y'' + 5y' + 6y = 0$, $y(0) = 1$, $y'(0) = 0$, and classify the damping.
- 21 For the general damped spring equation $my'' + cy' + ky = 0$ ($m, c, k > 0$), explain in one sentence why the real part of every root of the auxiliary equation is always negative (so every solution decays).

Verifying general solutions and the superposition principle

- 22 Verify by direct substitution that $y = C_1e^{2x} + C_2e^{3x}$ solves $y'' - 5y' + 6y = 0$ for any constants C_1, C_2 .
- 23 State the superposition principle for a linear homogeneous ODE, and explain why it guarantees that $y = C_1y_1 + C_2y_2$ solves $ay'' + by' + cy = 0$ whenever y_1, y_2 each do.
- 24 Find the auxiliary equation's roots for $y'' - 4y = 0$, then verify that both $y = e^{2x}$ and $y = e^{-2x}$ individually satisfy the ODE.

Boundary value problems

- 25 Solve the boundary value problem $y'' + y = 0$, $y(0) = 0$, $y(\pi/2) = 3$.
- 26 Solve $y'' - y = 0$, $y(0) = 1$, $y(1) = e$.

- 27 Explain why the boundary value problem $y'' + 4y = 0$, $y(0) = 0$, $y(\pi) = 1$ has NO solution, while $y(0) = 0$, $y(\pi/4) = 1$ does.